

Learning target 1.8, version 2

Find the equation for the tangent line to a function at a given point, and use tangent lines to approximate function values.

Suppose that for some function $p(x)$, you know the following information:

- $p(-2) = 5$,
- $p'(-2) = 1.5$,
- $p''(x) < 0$ for x -values close to -2 .

1. Explain and demonstrate how to find the linearization $L(x)$ of $p(x)$ at $x = -2$.

2. Explain and demonstrate how to estimate the value of $p(-2.03)$ using this linearization.

3. Is your estimate of $p(-2.03)$ greater than or less than the actual value? How do you know?

4. Sketch a possible graph of $p(x)$ and its linearization $L(x)$ nearby $x = -2$ to illustrate your findings. Label important points in your sketch with their coordinates.

Learning target 1.8, version 3

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The funny thing about square roots is that most of the time, they are *irrational* numbers, which means that they can't be written as fractions, and thus their decimal expansions go on forever and never repeat. For instance, $\sqrt{67} = 8.18535277187244996995370372473392945888048681549803996306671520$. That is clearly not a particularly useful answer to the question "what's $\sqrt{67}$?" We might in particular prefer to come up with a *fraction* that is a reasonably good approximation.

- (a) The number $\sqrt{67}$ is an output y -value of the function $f(x) = \sqrt{x}$. What's the corresponding input x -value?

- (b) Think of an x -value that's close to the input value you said in part (a), but for which it is *easy* to figure out the square root.

$$f(\text{____}) = \sqrt{\text{____}} = \text{____}$$

- (c) Compute $f'(x)$ and plug in your easy x -value. Leave your answer as a fraction.

$$f'(\text{____}) = \text{____}$$

- (d) Write down the equation for the tangent line to $f(x)$ at the easy x -value.

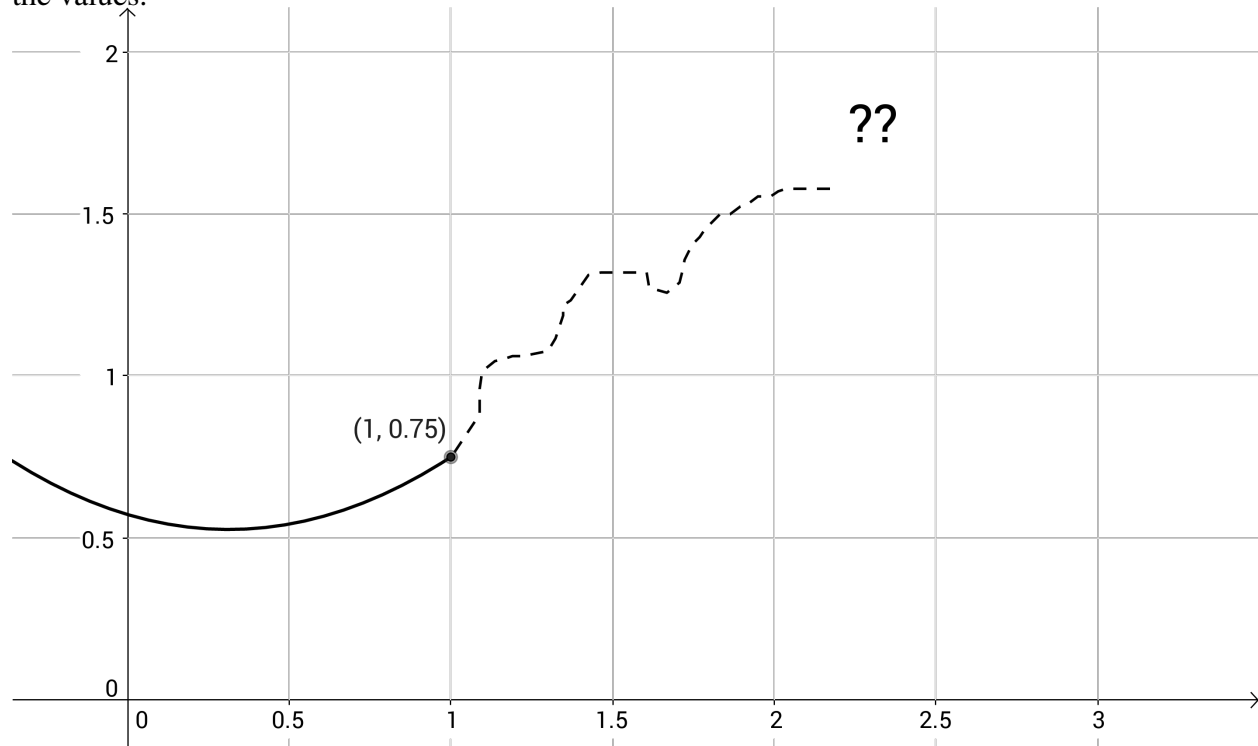
$$L(x) =$$

- (e) Lastly but not leastly, plug in the hard x -value into the equation for the tangent line. Leave your answer as a fraction.

Learning target 1.8, version 4

Find the equation for the tangent line to a function at a given point, and use tangent lines to approximate function values.

Part of the graph of this function $f(x)$ has been erased. We'll use linear approximation to recover the values.



- (a) Draw the tangent line to this function at $x = 1$.
- (b) Suppose we know that $f'(1) = 0.65$. Use point-slope form to write down an equation for $L(x)$, the tangent line to this function at $x = 1$.
- (c) Use your equation for $L(x)$ to estimate $f(1.4)$.
- (d) Do you think it is more likely that this is an overestimate or an underestimate? How come?